

Query-Conditioned Evidence Allocation for Efficient Retrieval-Augmented Generation

Project Specification and Locked Execution Plan v2.1 Short Name: QCCA (Query-Conditioned Context Allocator)

CS787 Graduate Research Project · 8-Week Timeline

Research Pair Programming Team

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Project Status: Locked Experimental Plan (v2.1)

Status: LOCKED EXPERIMENTAL SPECIFICATION (v2.1). This document serves as the authoritative methodological specification for the research project. The core Layer-1 QCCA study remains the primary research deliverable. A clearly separated Layer-2 experiment evaluates query-conditioned evidence selection without conflating budget allocation with passage identity. Any methodological alteration must be explicitly justified and documented before final-test evaluation.

1. Executive Summary & Core Research Question

Core Research Architecture: Two Experimental Layers

The project addresses adaptive context representation in RAG through two cleanly decomposed experimental layers:

Layer 1 — QCCA: HOW MUCH Evidence? *Does retrieval-side structure provide enough information to estimate how much natural-language evidence a query requires, and can a lightweight query-conditioned allocator dynamically choose this amount while preserving answer quality and reducing context cost?*

$$k(q) \in K_{\text{alloc}} = \{2, 4, 5, 6, 8\}$$

Top- $k(q)$ passages remain in full natural language; remaining $10 - k(q)$ passages use SARA's compressed representation. Layer 1 operates independently using standard BM25 ranking.

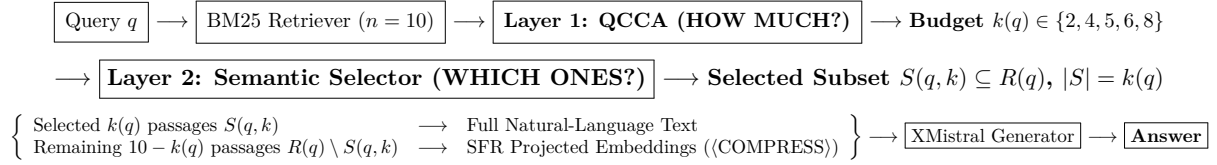
Layer 2 — Semantic/Topic Evidence Selector: WHICH Evidence? *Given a fixed evidence budget $k(q)$, can query-conditioned semantic/topic relevance identify which retrieved passages should receive the high-fidelity natural-language representation?*

$$S(q, k) \subseteq R(q), \quad |S(q, k)| = k(q)$$

This explicitly decouples **budget allocation** (HOW MANY?) from **evidence selection** (WHICH ONES?). Layer 2 is an additive extension; failure of Layer 2 does not compromise the core Layer-1 QCCA study.

Retriever	Compressor	Generator	Dataset	Primary Quality	Hardware
BM25 ($n = 10$)	SFR-Embedding-Mistral	Mistral-7B-Instruct-v0.2	QASPER	Token F1	NVIDIA TITAN RTX (24GB)

Two-Layer Conceptual Research Architecture:



Core Scientific Chain to Test:

1. Different queries exhibit heterogeneous sensitivity to context volume.
2. The optimal/acceptable evidence budget $k^*(q)$ therefore varies across queries.
3. Retrieval-side statistics contain predictive signal about this variation.
4. A lightweight allocator can exploit this signal before compression/generation.
5. Conditioned on predicted budget $k(q)$, semantic relevance may identify a superior subset of passages to preserve in natural language.
6. If successful, adaptive allocation improves the quality-efficiency tradeoff over the best static allocation.

A rigorous negative empirical result for either layer is scientifically valid and acceptable.

2. Research Motivation

The overarching research theme is **Query-Conditioned Resource Allocation**.

- **Prior Paradigm:** Internal Transformer computation allocation (predicting required Transformer layers or routing across mixture-of-experts).
- **Current Paradigm:** Evidence/context allocation prior to generation:
 - *Layer 1 (Adaptive Quantity):* Deciding how much uncompressed evidence must be fed into the generator versus dense compression.
 - *Layer 2 (Adaptive Identity):* Deciding which specific retrieved passages receive the uncompressed representation under that budget.

3. Research Hypotheses & Falsification Conditions

3.1 Layer 1: Allocation Hypotheses (Core)

H1: Query-Dependent Evidence Requirement

Different queries exhibit different sensitivities to the volume of natural-language evidence available to the generator. Consequently, the optimal/acceptable evidence budget $k^*(q)$ varies across queries.

H2: Retrieval Structure Contains Predictive Signal

Features derived from the retrieval-score distribution (top-1 concentration, top-1/top-2 margin, retrieval entropy, passage score counts, query length, and surface features) can predict the required natural-language evidence.

H3: Adaptive Allocation Improves Quality-Efficiency Tradeoff

A query-conditioned allocator achieves comparable or superior answer quality relative to a strong fixed- k baseline while reducing context token overhead. The critical controlled comparison is **QCCA vs. Best Static k** , not merely against parent SARA $k = 5$.

3.2 Layer 2: Evidence Selection Hypotheses (Secondary Extension)

H4: Query-Conditioned Evidence Selection

For a fixed evidence budget k , the BM25 top- k passages are not necessarily the passages that maximize answer-relevant evidence. A query-conditioned semantic/topic selection mechanism may identify a superior subset of the retrieved top-10.

H5: Layered Allocation Improves the Quality-Efficiency Tradeoff

Conditioned on the same predicted evidence budget $k(q)$, selecting WHICH passages remain uncompressed improves answer quality and evidence coverage relative to simply retaining the BM25 top- k .

3.3 Falsification Conditions

The project's hypotheses are falsified if any of the following occur:

- (a) Optimal k^* exhibits negligible variation across queries (a single static k dominates).
- (b) Retrieval-side features provide negligible predictive signal regarding evidence sufficiency.
- (c) QCCA fails to improve the quality-efficiency frontier relative to the best static allocation k_{dev}^* .
- (d) Layer-2 semantic selection yields no answer quality or evidence coverage gain over BM25 top- k .
- (e) Any apparent empirical advantage disappears under paper-level clustered bootstrap testing.

Note: Failure of Layer 2 (H4/H5) does not invalidate the primary Layer-1 QCCA study.

4. Scientific Problem Formulation

For query q , BM25 returns ranked passages $R(q) = \{p_1, p_2, \dots, p_{10}\}$ with scores $s_1 \geq s_2 \geq \dots \geq s_{10}$.

4.1 Candidate Evaluation & Allocation Spaces

We strictly distinguish two candidate sets:

- **Complete Sweep Grid:** $K_{\text{sweep}} = \{0, 2, 4, 5, 6, 8, 10\}$ used to characterize the full sensitivity curve ($k = 0$ pure compression floor; $k = 5$ parent SARA; $k = 10$ uncompressed vanilla RAG ceiling).
- **Deployable Allocator Action Space:** $K_{\text{alloc}} = \{2, 4, 5, 6, 8\}$. Note that $k = 5$ is explicitly included so QCCA can naturally reproduce the parent SARA configuration if appropriate.

4.2 Layer 1: Allocation Policy (HOW MUCH?)

$$k(q) = f(X(q)) \in K_{\text{alloc}} \quad (1)$$

where $X(q)$ contains strictly pre-generation retrieval features.

4.3 Layer 2: Selection Policy (WHICH ONES?)

Let $S(q, k) \subseteq R(q)$ with $|S(q, k)| = k$ denote the passages selected for natural-language representation.

- **Layer 1 Baseline Selection (BM25 Top- k):**

$$S_{\text{BM25}}(q, k) = \{p_1, p_2, \dots, p_k\} \quad (2)$$

- **Layer 2 Semantic/Topic Selection:**

$$S_{\text{semantic}}(q, k) = \text{TopK}_{p_i \in R(q)} \text{Rel}(q, p_i) \quad (3)$$

where $\text{Rel}(q, p_i)$ denotes a pre-generation query-passage semantic or topic relevance score.

4.4 Unified Context Representation

The generator context is constructed via:

$$\mathcal{C}_{k,S}(q) = [S(q, k)]_{\text{text}} \circ [\pi(\text{SFR}(p))]_{\langle \text{COMPRESS} \rangle, p \in R(q) \setminus S(q,k)} \quad (4)$$

The underlying SARA embedding projection and compression mechanism remains unchanged.

Strict Pre-Generation Invariant for Both Layers

$X(q)$ and $\text{Rel}(q, p_i)$ contain **strictly** information available prior to compression or generation. Neither layer may access gold answers, gold evidence spans, generated answers, downstream F1/ROUGE metrics, or human scores at inference time.

5. Primary Dataset & Scope Control

- **Benchmark:** QASPER (allenai/qasper), consisting of scientific paper QA over long full-text research articles with localized and multi-paragraph evidence annotations.
- **Scope Control:** The core 8-week CS787 research project focuses solely on QASPER. Cross-domain benchmarks remain post-course publication extensions.

6. Data Splits and Leakage Control

Critical Rule: Document-Level Disjointness

All partitioning is performed strictly at the **paper level** (`paper_id`), never at the question level. Questions from the same paper must never appear across different development/test partitions.

Partition Split	Approx. Papers	Approx. Questions	Designated Operational Scope
Discovery	~ 11	~ 35	Early sensitivity checks, code debugging, and exploratory distributions. Excluded from final allocator training.
Development	~ 90	~ 320	Fixed- k sweeps, k_{dev}^* selection, oracle construction, feature extraction, ϵ selection, GroupKFold validation, Layer-2 selector tuning.
Final Test	~ 180	~ 650	Frozen evaluation. Touched strictly once after all models, features, selectors, prompts, and code are locked.

Table 1: Paper-disjoint dataset partitions for the QCCA project.

7. Existing SARA Implementation & Hardware Constraints

- **Base Repository:** SARA-main.
- **Retriever:** BM25 through LlamaIndex, $n = 10$ passages, SentenceSplitter (`chunk_size = 256`, `overlap = 0`).

- **Generator:** mistralai/Mistral-7B-Instruct-v0.2 with XMistral 2-layer MLP projector.
- **Compressor:** Salesforce/SFR-Embedding-Mistral ($d = 4096$).
- **Special Configurations:** $k = 10$ (vanilla RAG, no compression), $k = 5$ (parent SARA), $k = 0$ (pure compression).

7.1 Hardware Constraint (NVIDIA TITAN RTX, 24GB VRAM)

Measured GPU memory requirements:

- Mistral-7B Generator + Projector: ~ 13.56 GB.
- SFR Compressor: ~ 13.24 GB.
- Combined footprint: $13.56 + 13.24 = 26.80$ GB > 24.0 GB (Causes CUDA Out-Of-Memory).

Execution Strategy: We execute sequential pipeline stages: (1) Load SFR, precompute/cache passage embeddings to disk, unload SFR; (2) Load Mistral/SARA, run generation using cached representations.

8. Baseline Hierarchy

8.1 Layer 1: Evidence Allocation Baselines

1. **Baseline 1 (Vanilla RAG, $k = 10$):** Full natural-language context; no compression.
2. **Baseline 2 (Pure Compression, $k = 0$):** All passages compressed; measures performance floor.
3. **Baseline 3 (Parent SARA, $k = 5$):** Official fixed allocation reproduction.
4. **Baseline 4 (Best Static SARA, $k = k_{\text{dev}}^*$):** Best static $k \in \{2, 4, 5, 6, 8\}$ selected on development data using the *same* quality-efficiency objective. **Primary controlled comparison.**
5. **Baseline 5 (Random Allocation):** Reviewer defense baseline sampling $k \sim \{2, 4, 5, 6, 8\}$ to verify that gains stem from intelligent allocation rather than random context variance.
6. **Related Work (RECOMP):** Conceptual related work only (uses FLAN-UL2 20B with non-native setup; avoids diluted reproduction effort).

8.2 Layer 2: Controlled Evidence Selection Baselines

Conditioned on the identical predicted evidence budget $k(q)$:

- (a) **QCCA + BM25 Top- k :** Standard Layer-1 system where $S(q, k) = \{p_1, \dots, p_k\}$.
- (b) **QCCA + Semantic Selector:** Layer-2 system where $S(q, k) = \text{TopK}_{p_i} \text{Rel}(q, p_i)$.
- (c) **Selection Oracle (Diagnostic):** Selects the subset $S^*(q, k) \subset R(q)$ that maximizes gold-evidence coverage from QASPER annotations. Offline theoretical diagnostic only.

The primary Layer-2 comparison is **QCCA + Semantic Selector vs. QCCA + BM25 Top- k** , isolating the value of answering WHICH passages receive verbatim representation under the exact same budget $k(q)$.

9. Fixed- k Experimental Matrix

Every development question is evaluated across the complete grid:

$$k \in K_{\text{sweep}} = \{0, 2, 4, 5, 6, 8, 10\} \quad (5)$$

Recording per question- k pair: Token F1, Exact Match (EM), ROUGE-L, input tokens, output tokens, compression latency (L_{comp}), generation latency (L_{gen}), selected k , retrieval scores, retrieval features, and peak VRAM. This produces the complete $|Q_{\text{dev}}| \times |K_{\text{sweep}}|$ evaluation matrix.

10. Primary Optimization Objective & Dual Oracles

10.1 Epsilon-Constrained Quality-Efficiency Formulation

For each question q , the minimum acceptable evidence budget is defined as:

$$k_{\epsilon}^*(q) = \min \left\{ k \in K_{\text{alloc}} \mid \text{F1}(q, k) \geq \max_{j \in K_{\text{sweep}}} \text{F1}(q, j) - \epsilon \right\} \quad (6)$$

Candidate values $\epsilon \in \{0.00, 0.01, 0.02, 0.05\}$ are evaluated on development data. Once selected, ϵ is frozen.

10.2 Two Diagnostic Oracles

- **Quality Oracle:** $k_{\text{qual}}^*(q) = \arg \max_{k \in K_{\text{alloc}}} \text{F1}(q, k)$ (ties broken to smaller k). Theoretical upper bound diagnostic.
- **Epsilon Oracle:** $k_{\epsilon}^*(q)$ as defined above. Efficiency-aware training target and benchmark.

Note: Oracles rely on observed post-generation quality and serve exclusively as offline diagnostics.

11. Oracle Gap Analysis

We quantify three critical performance gaps:

Oracle $\longrightarrow$ Best Static	[Amount of cross-query heterogeneity available for adaptive policies]
Oracle $\longrightarrow$ QCCA	[Remaining allocator optimization gap]
QCCA $\longrightarrow$ Best Static	[Empirical value of adaptive evidence allocation]

If Oracle $\approx$ Best Static, the data contains little heterogeneity for any allocator to exploit.

12. Evaluation Metrics & Evidence Quality Audit

- **Answer Quality:** Official QASPER token-level F1 (primary), ROUGE-L, Exact Match (EM).
- **Efficiency Metrics:** Input tokens, context reduction percentage, compression latency, generation latency, synthesized model-resident latency, and mean selected k .
- **Layer-2 Evidence-Selection Metrics (Offline Diagnostic):**
 - Gold evidence recall / coverage against annotated QASPER answer spans.
 - Selected-passages overlap percentage with gold evidence.
 - Full-evidence preservation fraction (queries whose selected k passages contain all necessary evidence).
- **Semantic Metric Audit:** Blinded human evaluation of 50 development questions at $k \in \{2, 5, 10\}$ scored as 0 = incorrect, 1 = partially correct, 2 = fully correct to verify correlation with Token F1.

13. Retrieval Features & BM25 Normalization

Given ranked BM25 scores $s_1 \geq s_2 \geq \dots \geq s_{10}$, we apply non-negative query-relative normalization:

$$s'_i = s_i - \min(s) + \varepsilon_{\text{num}} \quad (7)$$

Unit tests verify robustness against all-zero, equal, negative, and dominant score distributions.

13.1 Candidate Feature Set

1. **Top-1 Concentration:** $\rho_1 = \frac{s'_1}{\sum_{i=1}^{10} s'_i}$
2. **Top-1 / Top-2 Margin:** $\Delta_{12} = \frac{s'_1 - s'_2}{s'_1 + \epsilon_{\text{num}}}$
3. **Retrieval Entropy:** $p_i = \frac{\exp(s'_i/\tau)}{\sum_j \exp(s'_j/\tau)}$, $H = -\sum_{i=1}^{10} p_i \ln p_i$, $H_{\text{norm}} = \frac{H}{\ln(10)}$
4. **Query Length:** Number of tokens/words in query q .
5. **High-Score Passage Count (N_{high}):** Number of passages where $s'_i \geq \theta_{\text{rel}} \cdot s'_1$.
6. **Query Surface Category:** Heuristic indicator (Boolean, Factoid, Explanatory).
7. **Raw BM25 Score (s_1):** Optional ablation feature to test raw magnitude utility.

14. QCCA Architecture & Layer-2 Selector Design

14.1 Layer 1: QCCA Allocator (HOW MUCH?)

Candidate allocation space: $k(q) \in K_{\text{alloc}} = \{2, 4, 5, 6, 8\}$.

- **QCCA-Rule:** Piecewise policy mapping high concentration ($\rho_1 \geq \theta_{\text{high}}$) to smaller k ($k = 2$), moderate concentration to $k \in \{4, 5, 6\}$, and diffuse retrieval to $k = 8$. Thresholds learned strictly on development data.
- **QCCA-Learned:** L2-regularized multinomial Logistic Regression (primary) and shallow Decision Tree (secondary), targeting k_{ϵ}^* .
- **GroupKFold Validation:** 5-fold cross-validation grouped strictly by `paper_id` to prevent document-level information leakage.

14.2 Layer 2: Topic/Semantic Evidence Selector Design (WHICH ONES?)

Methodological Note on Layer-2 Implementation

The Layer-2 selector will be implemented using a lightweight semantic/topic relevance mechanism appropriate to the retrieved passage/query setting. The exact method will be selected on development data and frozen before final-test evaluation. The selector must use only pre-generation information and must not use gold answers, gold evidence annotations, generated answers, F1/ROUGE, or human judgments at inference time.

Exploratory Development Alternatives:

1. *Query-Passage Embedding Cosine Similarity:* Scoring passages via cosine similarity against the query using SFR embeddings already resident or cached.
2. *Lightweight Topic/Semantic Representation:* Assessing topic alignment between query terms and passage representations.
3. *Clustering/Topic Coverage:* Selecting passages that maximize semantic coverage and minimize redundant overlap across the retrieved top-10.

Controlled Layer-2 Execution Flow: For every evaluation question:

1. Retrieve top-10 passages using BM25.
2. Predict evidence budget $k(q) \in \{2, 4, 5, 6, 8\}$ using frozen Layer-1 QCCA.
3. *Branch A (Layer-1 standard):* Retain top- $k(q)$ BM25 passages as natural language.
4. *Branch B (Layer-2 selector):* Score top-10 passages with the frozen semantic selector; retain top- $k(q)$ scoring passages as natural language.
5. Both branches compress remaining $10 - k(q)$ passages using identical SFR embeddings, feed the context into Mistral-7B with identical prompts and decoding parameters, and evaluate answer quality.

15. Retrieval-Ceiling Analysis

To avoid conflating retrieval, allocation, and selection failures, we partition errors:

- **Retrieval Failure:** Golden evidence is absent from the top-10 retrieved set (retriever-bounded).
- **Allocation Failure (Layer 1):** Golden evidence was retrieved, but QCCA selected an insufficient budget k , forcing key evidence into compressed form.
- **Selection Failure (Layer 2):** Golden evidence was in the top-10 and budget k was sufficient, but the selector omitted key evidence passages from $S(q, k)$.
- **Generation Failure:** Relevant evidence was presented in natural language, but the generator produced an incorrect answer.

Performance is reported separately on retrieval-success and retrieval-failure subsets.

16. Latency Methodology: Synthesized Model-Resident Latency

To prevent hardware sequential swapping from distorting algorithmic benchmarking:

- Measure $L_{\text{comp}}(k)$ with SFR pre-loaded.
- Measure $L_{\text{gen}}(k)$ with Mistral pre-loaded.
- Measure selector latency $L_{\text{select}}(k)$ for Layer 2.
- Compute **Synthesized Model-Resident Latency**:

$$L_{\text{total}}(q) = L_{\text{select}}(q) + L_{\text{comp}}(10 - k(q)) + L_{\text{gen}}(k(q)) \quad (8)$$

Excludes model loading/unloading, disk I/O, and sequential PCIe overheads.

17. Master 8-Week Execution Timeline

Week	Phase Focus	Core Engineering & Scientific Deliverables
W1	Infrastructure	Freeze Python/library versions; reproduce SARA unit tests and inference; verify sequential GPU memory; generate paper-level split manifests; build feature extractor and SFR caching harness. [Go/No-Go Gate]
W2	Dev Fixed- k Matrix	Run complete $k \in K_{\text{sweep}}$ sweep on Development set; log quality, token, and latency metrics; establish Vanilla RAG, Pure Compression, and Parent SARA baselines.
W3	Oracle & Target Dev	Compute k_{qual}^* and k_{ϵ}^* ; analyze optimal k distributions; verify hypothesis H1; conduct blinded 50-question human audit; select and freeze ϵ^* on development data.
W4	QCCA Development	Train QCCA-Rule and QCCA-Learned using GroupK-Fold; analyze feature importance; execute full dry-run of final test harness using a dummy random allocator.
W5	Methodology Freeze & QCCA Test	LOCK ALL features, models, ϵ^* , thresholds, static baseline k_{dev}^* , prompts, and code. Execute Final Test set evaluation for Layer-1 QCCA systems.
W6	Layer-2 Dev & Controlled Eval	Implement and tune Layer-2 semantic/topic evidence selector on development split; benchmark against BM25 top- k under frozen QCCA budgets.
W7	Ablations, Failures & Statistics	Execute paper-clustered bootstrap ($B = 10,000$ re-samples) for 95% CIs; Pareto analysis (F1 vs. tokens/latency); run 3-tier ablations and deterministic failure analysis.
W8	Manuscript & Wrap-up	Finalize publication-ready LaTeX tables and Pareto figures; complete CS787 final report, presentation slides, and open-source reproducibility artifacts.

Table 2: Chronological 8-week operational milestone plan.

18. Final Test Evaluation Matrix

System Arm	Retriever	Selection	Generator	Allocation k	Experimental Role
<i>Layer 1: Allocation Baselines & Primary QCCA Arms</i>					
1. Vanilla RAG	BM25	Top- k	Mistral-7B	10	Full natural-language ceiling
2. Pure Compression	BM25	None	Mistral-7B	0	Compression performance floor
3. Parent SARA	BM25	Top- k	Mistral-7B	5	Fixed parent architecture
4. Best Static SARA	BM25	Top- k	Mistral-7B	k_{dev}^*	Primary static control baseline
5. Random Allocation	BM25	Top- k	Mistral-7B	$k \sim K_{\text{alloc}}$	Allocation variance control
6. QCCA-Rule	BM25	Top- k	Mistral-7B	$k(q)$	Interpretable rule policy
7. QCCA-Learned	BM25	Top- k	Mistral-7B	$k(q)$	Primary learned ML policy
8. Quality Oracle	BM25	Top- k	Mistral-7B	$k_{\text{qual}}^*(q)$	Theoretical quality ceiling
9. Epsilon Oracle	BM25	Top- k	Mistral-7B	$k_{\epsilon}^*(q)$	Efficiency-aware upper bound
<i>Layer 2: Controlled Evidence Selection Extension</i>					
10. QCCA + Semantic Selector	BM25	Semantic	Mistral-7B	$k(q)$	Evaluates WHICH evidence under budget $k(q)$
11. Selection Oracle	BM25	Oracle	Mistral-7B	$k(q)$	Offline diagnostic maximizing gold coverage

Table 3: Master system comparison matrix on the Final Test split.

19. Reporting Schema & Statistical Inference

Main Results Schema: System | Token F1 | ROUGE-L | EM | Input Tokens | Context Reduction % | Mean k | Evidence Coverage | L_{comp} | L_{gen} | L_{total} | Peak VRAM.

- **Paper-Clustered Bootstrap:** $B = 10,000$ iterations. Papers are sampled with replacement, evaluating all constituent questions to obtain paired difference distributions:

$$\Delta = \text{Metric}(\text{Candidate}) - \text{Metric}(\text{Baseline})$$

with 95% empirical confidence intervals.

- **Primary Controlled Comparisons:**
 - *Layer 1:* QCCA-Learned vs. Best Static SARA k_{dev}^* .
 - *Layer 2:* QCCA + Semantic Selector vs. QCCA + BM25 Top- k (identical predicted $k(q)$).
- **Non-Parametric Validation:** Wilcoxon signed-rank test on paired per-question differences.

20. Pareto Analysis & Ablation Matrix

- **Primary Pareto Frontier:** Context Input Tokens (X) vs. Token F1 (Y).
- **Secondary Pareto Frontier:** Synthesized Model-Resident Latency (X) vs. Token F1 (Y).

Ablation Hierarchy:

- **Tier 1 (Required):** Static- k sensitivity ($k \in K_{\text{sweep}}$), QCCA vs. Best Static, Oracle gap, Pareto analysis.
- **Tier 2 (Important):** Layer-2 Semantic Selector vs. BM25 Top- k , concentration-only features vs. full feature set, leave-one-feature-out, random allocation control, query-type breakdown.
- **Tier 3 (Optional):** Semantic selector variant ablations (cosine vs. topic vs. clustering), entropy temperature (τ) sensitivity.

21. Deterministic Failure Analysis Taxonomy

Failure cases are identified via quantitative rules and classified into five categories:

1. **Over-Compression:** Selected k is smaller than k_e^* ; relevant evidence was excessively compressed.
2. **Under-Compression:** Selected k is larger than necessary, inflating context tokens without quality gain.
3. **Selection Failure:** Budget $k(q)$ was adequate, but semantic selector prioritized distractor passages over gold evidence.
4. **Retrieval Failure:** Golden evidence absent from top-10 passages (retriever-bounded).
5. **Generation Failure:** Relevant evidence was uncompressed, but Mistral-7B produced an incorrect answer.

22. Project Risk Matrix & Guardrails

Risk Factor	Root Cause / Mechanism	Engineering & Methodological Mitigation
GPU OOM	Concurrent SFR + Mistral footprint (26.80 GB) exceeds 24GB VRAM.	Enforce two-stage sequential execution; precompute and cache passage embeddings to disk.
Allocator Overfitting	Small dev set (~ 320 questions) with expressive models.	Enforce GroupKFold by paper; apply strong L2 regularization; restrict decision trees to shallow depth.
Best Static $\approx$ QCCA	Retrieval statistics contain insufficient predictive signal.	Valid negative scientific finding. Report honestly, document oracle gap, and characterize limits.
Layer-2 Instability	Passage selector is noisy or adds latency without quality gains.	Layer 2 is an additive extension; Layer-1 QCCA remains the guaranteed, independent deliverable.
Data Leakage	Cross-contamination of paper topics across splits.	Strict paper-level partitioning verified by automated manifest assertions.
Metric Noise	Generative scientific QA causes noise in string metrics.	Conduct blinded 50-question human audit; correlate with Token F1.

Table 4: Identified technical risks and engineering mitigations.

Project Guardrails: Prohibited Actions

Do **not**: fine-tune retriever or generator; introduce additional heavy model families; tune on final-test data; alter prompts after test evaluation begins; use post-generation information in $X(q)$ or $\text{Rel}(q, p)$; use gold evidence for passage selection at inference time; alter the QCCA allocator during Layer-2 evaluation; or allow Layer-2 development to retroactively modify frozen Layer-1 checkpoints.

23. Summary & Definitive Benchmark Standard

Final One-Sentence Project Statement

Instead of giving every RAG query the same amount of full-context evidence, QCCA tests whether inexpensive retrieval-side signals can determine how much natural-language evidence a query requires; a secondary Layer-2 experiment then tests whether, given that evidence budget, query-conditioned semantic selection can determine which retrieved passages should receive the high-fidelity representation.